Simplification of Nested Radicals

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Extended Abstract

Abstract

Radical simplification is a fundamental mathematical question as well as an important part of symbolic computation systems. The general denesting problem had not been known to be decidable. We give necessary and sufficient conditions for a radical α over a field k to be denested. We give the first algorithm to decide if the expression can be denested. This algorithm computes an equivalent expression of minimum nesting depth. The algorithm has running time polynomial in the size of the splitting field of the minimal polynomial of α over k.

Introduction

In his magic way, Ramanujan observed a number of striking relationships between certain nested radicals:

$$\sqrt[3]{\sqrt[3]{2}-1} = \sqrt[3]{1/9}-\sqrt[3]{2/9}+\sqrt[3]{4/9} \quad (1)$$

$$\sqrt{\sqrt[3]{5}-\sqrt[3]{4}} = 1/3(\sqrt[3]{2}+\sqrt[3]{20}-\sqrt[3]{25}) \quad (2)$$

$$\sqrt[6]{7\sqrt[3]{20}-19} = \sqrt[3]{5/3}-\sqrt[3]{2/3} \quad (3)$$

These remained innocent curiosities for decades. Then symbolic computation came of age, and such manipulations assumed greater importance. The equation:

$$\sqrt{5+2\sqrt{6}}=\sqrt{2}+\sqrt{3}$$

means that $\{1, \sqrt{2}, \sqrt{3}, \sqrt{6}\}$ is a basis for $Q(\sqrt{5+2\sqrt{6}})$ over Q. It is much simpler to manipulate than the basis consisting of powers of $\sqrt{5+2\sqrt{6}}$. Simplification of nested radicals is important for computations in dynamical systems. Thus the "denesting" of radicals – a term which will be precisely defined in the next section – is quite important for symbolic computation systems as well as being of fundamental mathematical interest.

In 1984, Borodin, Fagin, Hopcroft and Tompa [3] gave an efficient algorithm for decreasing the nesting depth of a class of expressions involving square roots. Zippel [17] gave some conditions under which a radical could denest. The general case remained open.

We show that a radical can be denested iff the denesting occurs within its splitting field. That is, a radical α can be denested over a field k iff the denesting occurs with each term of the denesting lying within the splitting field of the minimal polynomial of α over k. Thus the problem is decidable, in marked contrast to the simplification problem for expressions involving trigonometric and exponential functions [11]. Following the measure of the size of univariate polynomials in the factoring problem, we define the size of the denesting problem to be the minimal polynomial for α over k. (See §3 for details.) Then we show that given a radical α over a field k of characteristic 0, we can denest α in the time it takes to compute the splitting field of the minimal polynomial of α over k. In worst case the splitting field can be exponentially large.

Our paper is organized as follows: §1: Background, §2 Algebraic Structure, §3 The Algorithm and Running Time Analysis, and §4 Conclusions. We omit proofs in this extended abstract.

1 Background

We begin with a brief review of some algebraic concepts. Let k be a field. Throughout this paper we assume that k is of characteristic 0. An element α *is algebraic over* k iff α satisfies a polynomial with coefficients in k. The *degree* of α is the degree of the minimal irreducible polynomial of α over k. If L has finite dimension $[L:k]$ over a subfield k, then $L=k(\theta)$ for some θ in L, where the degree of θ over k equals $[L:k]$.

An extension field L is *algebraic over a field* k iff every

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element of L is algebraic over k. It is well known that every finite extension of a field is algebraic; the finite extensions of Q are the called the *algebraic number fields.* If k is an algebraic number field, it contains a set of elements which satisfy integer monic polynomials over Z; this is called the *ring of integers of* k, and is frequently denoted O_k.

In 1982, Lenstra, Lenstra and Lovász [9] showed that $f(x)$ in $Q[x]$ can be factored in polynomial time. If $f(x)$ is a polynomial in O_k, where $k = Q[t]/g(t)$ is a number field, then there are polynomial time algorithms for factoring $f(x)$ over O_k [6], [8].

Following [3], a *formula* over a field k and its *depth of nesting* are defined as follows:
(1) an element of k is a formula of depth 0 over k,
(2) an arithmetic combination (A+B, A $\times$ B, A/B) of formulas A and B is a formula whose depth over k is max(depth(A), depth(B)), and
(3) a root $\sqrt[n]{A}$ of a formula A is a formula whose depth over k is 1+ depth(A).

We will say the formula A *can be denested over the field* k if there is a formula B of lower depth than A such that value(A)=value(B). For any α, we define the depth of α over k to be $minimum\{depth(A) \mid value(A) = \alpha\}$. We will write a primitive n^{th} root of unity as a special symbol ξ_n, rather than as a nested radical.

We return to the examples mentioned earlier. At first they seem mysterious. In fact, each equation is a result of a complex algebraic structure. Consider:

$$\sqrt[3]{\sqrt[3]{2}-1} = \sqrt[3]{1/9} - \sqrt[3]{2/9} + \sqrt[3]{4/9}.$$

A natural question to ask is: What is the relationship between $\sqrt[3]{2}$ and $\sqrt[3]{9}$? There is no obvious one. But there is a relationship between $\sqrt[3]{\sqrt[3]{2}-1}$ and $\sqrt[3]{9}$; $\sqrt[3]{9}$ is an element of the field $Q(\sqrt[3]{\sqrt[3]{2}-1})$. This fact is unexpected.

Equations 2 and 3 give similar results. A natural questions arises: In which field do denestings occur?

Let k be an algebraic number field, and let $f(x)$ be an irreducible polynomial of degree n with coefficients in k, and roots $\alpha_1, \ldots, \alpha_n$. Then $k(\alpha_i) \simeq k[x]/f(x) \simeq k(\alpha_j)$, but in general $k(\alpha_i) \neq k(\alpha_j)$. The field $L = k(\alpha_1, \ldots, \alpha_n)$ is called the *splitting field of f(x) over k*. This is the smallest field containing $k(\alpha_1)$ that is Galois over k[1] We consider the set of automorphisms of L which leave k fixed. These form a group, called the *Galois group of* L over k. As we can think of these automorphisms as permutations of the α_i, this group is sometimes called the *Galois group of f(x) over k*. If $f(x)$ is irreducible over k, the Galois group is *transitive* on $\{\alpha_1, \ldots, \alpha_n\}$, that is, for each pair of elements α_i, α_j, there is an element σ in G, with $\sigma(\alpha_i) = \alpha_j$. With remarkable insight, Galois discovered a relationship between the subgroups of G and the subfields of L containing k.

[1] We say a field $F \supseteq k$ is *Galois* over k if every irreducible polynomial $p(x)$ in $k[x]$ that has a root in F splits completely in F.

Let H be a subgroup of G. We denote by L^H the set of elements of L which are fixed pointwise by each element of H. This set forms a field. Furthermore, H fixes k, so that we have:

$$k \subseteq L^H \subseteq L.$$

Conversely suppose that $J = k(\beta_1, \ldots, \beta_r)$ is a field such that $k \subset J \subset L$. Then the β_i can be written as polynomials in $\alpha_1, \ldots, \alpha_n$, and H, the subgroup of G which fixes J, consists of those elements of G which fix the β_i pointwise. The relationship between the fields and the groups can be formally stated as:

Theorem 1.1 *Fundamental Theorem of Galois Theory: Let k be a field, and let $f(x)$ in $k[x]$ be a polynomial of degree n, with roots $\alpha_1, \ldots \alpha_n$. Then:*

1. *Every intermediate field J, with $k \subset J \subset L$ defines a subgroup H of the Galois group G, namely the set of automorphisms of L which leave J fixed. Furthermore, L is Galois over J.*
2. *The field J is uniquely determined by H, for J is the set of elements of L which are invariant under the action of H.*
3. *The subgroup H is normal iff J over k is a Galois extension. In that case the Galois group of J over k is G/H.*
4. *$|G| = [L : k]$, and $|H| = [L : J]$.*

We are now ready to state our main result:

Theorem 2.6 *Suppose α is a nested radical over k, where k is a field of characteristic 0. If α can be denested over k, then there is a denesting of α such that each of its terms lies in the splitting field of the minimal polynomial of α over k.*

2 Algebraic Structure

In this section we discuss the algebraic structure surrounding nested radicals. Essentially a denesting of α of depth l means that there is a tower of fields over k which are abelian extensions, that is to say, each extension is Galois, and each resulting Galois group is abelian. Thus we have:

Lemma 2.1 *Let α be a nested radical, and suppose α can be denested in a field $L \supseteq k$. Suppose that the denested expression has nesting depth l. Let $\tilde{L}$ be the splitting field*[2] *of L over k, with Galois group G, and assume that all roots of unity of $\tilde{L}$ appear in k. Then there are subgroups $H_1, \ldots, H_l$ of G, with $G = H_0 \rhd H_1 \rhd \ldots \rhd H_l$ and H_i/H_{i+1} abelian for $i = 0, \ldots l-1$, with $H \supseteq H_l$, where H is the subgroup of G associated with $k(\alpha)$. Conversely, if there is such a sequence of subgroups, then α has nesting depth at most l.*

We are now ready to proceed with:

Theorem 2.2 *Suppose α is a nested radical over k, a field of characteristic 0, and suppose further that all the roots of unity which lie in the splitting field of the minimal polynomial of α over k actually lie in k. If α can be denested over k, then there is a denesting of α such that each of its terms lies in the splitting field of the minimal polynomial of α over k.*

There are two pieces left in the puzzle: (1) we must find a minimal length chain of subgroups H_i satisfying the conditions of Lemma 2.1, and (2) we must figure out how to eliminate the requirement that the roots of unity lie in the base field. Neither of these present a a problem.

We call the subgroup $G' = \langle ghg^{-1}h^{-1} \mid g, h \in G \rangle$ the *commutator subgroup of* G. The following is well-known:

Theorem 2.3 *Let G be a group with subgroup H. If $H_1, \ldots, H_l$ satisfy $G = H_0 \rhd H_1 \rhd \ldots \rhd H_l$, with H_i/H_{i+1} abelian, and $H_l \subset H$, then it is also true that $G = G^{(0)} \rhd G^{(1)} \rhd \ldots \rhd G^{(l)}$, and $G^{(l)} \subseteq H$ with $G^{(i)}/G^{(i+1)}$ abelian, where $G^{(i+1)} = G^{(i)\prime}$.*

The roots of unity present no difficulty either.

Lemma 2.4 *Let L be a Galois extension of k, with group G, and supoose ξ_s is in L. If $H = G'$ is the commutator subgroup of G, then $L^H \supseteq k(\xi_s)$.*

At this point, it becomes clear that we can remove the requirement in Lemma 2.1 and Theorem 2.2 that the needed roots of unity lie in k. They will be adjoined by the first commutator subgroup of G, and thus we have:

Theorem 2.5 *Suppose α, a nested radical over k, has minimal depth l. Let L be the splitting field of the minimal polynomial of α over k with Galois group G, and let H be the subgroup of G fixing $k(\alpha)$. Then $H \supseteq G^{(l)}$, but $H \not\supseteq G^{(l-1)}$.*

[2] This is usually called the normal closure of L over k, i.e. the minimal field $\tilde{L}$ containing L in which every polynomial in $k[x]$ which has a root in L splits completely in $\tilde{L}$.

Theorem 2.6 *Suppose α is a nested radical over k, where k is a field of characteristic 0. If α can be denested over k, then there is a denesting of α such that each of its terms lies in the splitting field of the minimal polynomial of α over k.*

3 The Algorithm and Running Time Analysis

In order to find a denesting of α over k, we begin by constructing the splitting field of the minimal polynomial of α over k. Using the standard techniques of Galois theory, we transform a question about polynomials to a question about groups. In Theorem 2.3 we have characterized the chain of subgroups which leads to a denesting of minimal depth. What we do now is show how to construct the appropriate field, given the subgroup. Finally we show how to realize that field as a radical extension. If $H_1, \ldots, H_l$ is the chain of subgroups associated with a denesting of α, and L is the splitting field of α over k, then $k(\alpha) \subseteq L^{H_l}$, and the field L^{H_l} contains a denesting of α which is of minimal depth l.

We first show how to construct the splitting field L of $k(\alpha)$ over k. Following Artin, we construct a polynomial $s(x)$ whose roots $\theta_1, \ldots, \theta_r$ form a "normal" basis for L over k; that is, degree $s(x) = [L:k]$, and $\theta_1, \ldots, \theta_r$ are linearly independent over k. This enables us to easily construct the fixed fields of L corresponding to subgroups of the Galois group G: if $L^H = k(\beta_1, \ldots, \beta_k)$, then the β_j are linear combinations of the θ_i. By Lemma 2.4, all the roots of unity appearing in L appear in $K = L^{G'}$, the field fixed by the commutator subgroup of G. Thus we can add them "first" without penalty.

From a sequence of subgroups H_i with $G = H_o \rhd H_1 \rhd \ldots \rhd H_f$, H_i/H_{i+1} cyclic, and $H_f \supseteq H$, where H is the subgroup of G belonging to $k(\alpha)$, we use Hilbert's Theorem 90 to construct a tower of fields $K \subset K(\beta_1) \subset \ldots \subset K(\beta_1, \ldots, \beta_f)$, where each extension is cyclic, and the expression for α in $K(\beta_1, \ldots, \beta_f)$ is of minimal nesting depth l. In our construction we will actually have two descriptions of the fields $K(\beta_1, \ldots, \beta_i)$, namely $K(\beta_1, \ldots, \beta_i)$ and $K[x]/g_i(x)$, where $g_i(x)$ is irreducible. We will do computations using the latter representation.

From an algebraic point of view, a sensible measure of the size of the denesting problem is the size of the minimal polynomial of α. In the full paper we show how to go from an expression of α of the form:

$$\begin{aligned} \alpha_1 &= \sqrt[n_1]{\tilde{p}_1}, \ \tilde{p}_1 \in k \\ \alpha_2 &= \sqrt[n_2]{\tilde{p}_2(\alpha_1)} \\ \alpha_3 &= \sqrt[n_3]{\tilde{p}_3(\alpha_1, \alpha_2)} \end{aligned}$$

$$\vdots$$
$$\alpha = \alpha_m = \tilde{q}(\alpha_1, \ldots, \alpha_{m-1}) + \sqrt[n_m]{\tilde{p}_m(\alpha_1, \ldots, \alpha_{m-1})},$$

to the minimal polynomial of $\alpha = \alpha_m$ over k. This polynomial is polynomial size in $n = \Pi n_i$, $||\alpha_i||$, $||\tilde{q}(x_1, \ldots, x_{m-1})||$ and $||\tilde{p}_i(x_1, \ldots, x_{i-1})||$. Whether or not the minimal polynomial is polynomial in the length of the expression for α is an open question.

Next we show that all the computations described above can be carried out in time polynomial in the size of the splitting field of the minimal polynomial of α over k. Our bounds are chosen for the relative simplicity of presentation, and are not necessarily the tightest possible. We will restrict our attention to $k = Q$ for the present.

In [6], it was observed that the splitting field and Galois group G can be computed in time polynomial in $r = |G| = [L : k]$ and $\log|f(x)|$. More precisely:

Theorem 3.1 *Let $f(x)$, an irreducible polynomial of degree m over k, be the minimal polynomial of α, a nested radical over k. The Galois group of $f(x)$ over k can be computed in $O(r^{15+\epsilon} m^{7+\epsilon} \log^{4+\epsilon}(r|f(y)|))$ steps.*

Thus in time polynomial in the degree of the splitting field, and the size of the minimal polynomial of α over k, we can compute the Galois group. It has long been known how to compute commutator subgroups quickly from a group table; recently Babai, Luks and Seress [2] showed how to do so in $O(r^4 \log^c r)$ steps, for a permutation group on r elements. This allows us to quickly compute the group H_l such that $L^{H_l} \supset k(\alpha)$ with α having an expression of minimal nesting depth in L^{H_l}. It now remains to compute the appropriate fields.

We say a polynomial $g(x)$ over a field k is normal iff it is irreducible and $k[x]/g(x)$ is a Galois extension of k (equivalently $g(y)$ splits completely in $k[x]/g(x)$). It is not hard to go from such a polynomial to one whose roots form a normal basis. Artin gave an effective method for calculating such a polynomial. His algorithm does so in $O(r^7 \log^{2+\epsilon} |f(y)|)$ steps.

We are now in a position to compute fixed fields. Without loss of generality, suppose $\{\sigma_1, \ldots, \sigma_l\}$ are the elements of H. Let $\beta = \Sigma a_j \theta_j$, with a_j in k, be an arbitrary element of L^H. For each σ_i in H, we have

$$\Sigma a_j \theta_j = \sigma_i(\Sigma a_j \theta_j) = \Sigma a_j \sigma_i(\theta_j). \tag{4}$$

Since the Galois group sends roots of $s(x)$ to roots of $s(x)$, we have $\sigma_i(\theta_j) = \theta_l$ for some l. Because the θ_j are linearly independent over k, the only way equation 4 can be satisfied is if $a_i = a_l$. Every element in H gives rise to equalities amongst the a_i, which leads to a series of relationships amongst the θ_i. That is, if $a_i = a_l$, then in L^H any expression containing θ_i must have θ_l appearing with the same coefficient. Computing all such relationships gives us exactly the fixed field associated with H. The full paper gives the algorithm.

Next we need to transform our extensions into extensions by radicals. We transform our tower of groups into one in which each extension is cyclic. This is of course easy. If $H_1, \ldots H_l$ is a series of subgroups of $G = H_0$ satisfying $H_i \triangleright H_{i+1}$ and H_i/H_{i+1} is abelian, then all we need to do is for each i, $i = 0, \ldots, l-1$, find subgroups $H_{i1}, \ldots, H_{ij_i}$, with H_i/H_{i1} cyclic, and H_{ik}/H_{ik+1} cyclic for $k = 1, \ldots, j_i - 1$. This can be done in time $O(r^4 \log^c r)$ steps [2].

We have shown how to compute $K_i = L^{H_i}$. We wish to show how to write K_i as a cyclic extension of K. Let us renumber the H_i so that $G \triangleright H_0 \triangleright H_1 \triangleright \ldots \triangleright H_f$, with H_i/H_{i+1} cyclic for $i = 0, \ldots, f-1$.

Suppose $[K_1 : K] = l_1$; then we know that ξ_{l_1} is in K. Thus the hypothesis of Hilbert's Theorem 90 are satisfied. Let σ be a generator of H_1. Then we seek an element β in K_1 such that $\xi = \beta/\sigma(\beta)$. If $K_1 = K(\beta_1, \ldots, \beta_n)$, then $\beta = \Sigma b_i \beta_i$, with the b_i in K. We know that

$$\xi = \frac{\Sigma b_i \beta_i}{\Sigma b_i \sigma(\beta_i)}.$$

We can solve for the b_i using linear algebra. (There may be more than one β, but that does not present a problem.) Once we have found a β, we observe that $\sigma^i(\beta) = \xi^i \beta$ for $i = 1, \ldots, l_1$. Since the elements $\xi^i \beta$ are distinct, we know that $[K(\beta) : K] \geq l_1$. Thus $K_1 = K(\beta)$. Finally we know that

$$\sigma(\beta^{l_1}) = (\sigma(\beta))^{l_1} = (\xi\beta)^{l_1} = \xi^{l_1}\beta^{l_1} = \beta^{l_1}.$$

This implies that β^{l_1} is in K. Let $b = \beta^{l_1}$; then β satisfies $x^{l_1} - b$.

There was nothing special about the fact that the extension was over K. The same construction will work over K_1, or any other field containing the appropriate roots of unity. Since all the needed roots of unity appear in K, that is not a problem.

Theorem 3.2 *On input $f(x)$, the minimal polynomial for α, a nested radical over k, the algorithm* **Compute Denesting** *computes a denesting of α of minimal nesting depth. It does so in the time required to compute the splitting field of the minimal polynomial of α over k.*

Algorithm: Compute Denesting

input: $f(x)$, an irreducible polynomial of degree n over k.
output: A sequence of fields K_i, $i = 1, \ldots, f$, and an expression for α, where

1. $K_r = L$ is the splitting field of $f(x)$ over k,
2. K_1 is the smallest subfield of L containing k which contains all the roots of unity of L,
3. K_{i+1} is a cyclic extension of K_i, and
4. the expression for α, a root of $f(x)$, in K_f is of minimal nesting depth.

Line 1: BEGIN
Line 2: Compute L, the splitting field of $f(x)$ over k;
Line 3: Compute K_1;
Line 4: Compute $G =$ Galois group of L over K_1;
Line 5: $H_0 \leftarrow G$;
Line 6: $H \leftarrow$ Galois group of L over $k(\alpha)$;
Line 7: $i \leftarrow 0$;
Line 8: WHILE $H_i \not\supseteq H$ DO:
Line 9: BEGIN
Line 10: $H_{i+1} \leftarrow H_i'$;
Line 11: $i \leftarrow i + 1$;
Line 12: END;
Line 13: Transform the sequence of groups H_i to a sequence of groups $J_j, j = 1 \ldots f$, in which each factor group is cyclic;
Line 14: For each group J_j DO:
Line 15: BEGIN
Line 16: Compute $K_j = L^{J_j} = K_{j-1}(\beta_1, \ldots, \beta_n)$;
Line 17: Pick σ, a generator of J_j;
Line 18: Solve for $\beta = \Sigma b_j \beta_j$, with b_j in K_{j-1}, $\xi_{l_j} = \frac{\Sigma b_j \beta_j}{\Sigma b_j \sigma(\beta_j)}$; (where ξ_{l_j} is a primitive l_j^{th} root of unity, and $[K_{j+1} : K_j] = l_j$)
Line 19: $K_{j+1} \leftarrow K_j(\beta)$;
Line 20: END;
Line 21: Express α in K_f;
Line 22: END.

We have presented our algorithm as an algorithm over Q. The only reason for that is simplicity. The only requirement for the base field is that one needs to be able to factor polynomials over k. We can generalize **Algorithm Compute Polynomial** to compute a minimal polynomial for α over k. Of course, we can generalize Theorem 3.1 , and **Algorithms Compute Fixed Field** and **Compute Denesting**.

4 Conclusions

In one sense, we have settled the question of denesting radicals, in another, we have not. We have given a structure theorem which completely categorizes when a nested radical can be denested, and we have given an algorithm to compute the denesting.

Our main algorithm is not fast. Thus we see the most important issue in this problem is speed. To this we have several comments:

1. Is there a way to perform these computations via a straightline program? The encoding in the straightline version would avoid the problem of coefficient blow-up. The main difficulty seems to be in computing the Galois group. We do not see how to determine the group without determining both $f(x)$, the minimal polynomial of α over k, and the splitting field of $f(x)$ over k. If a way could be found around these problems, then the entire algorithm could be sped up. We do not think this will be easy. In particular, it is likely that any technique which works here will also work to determine the Galois group in the solvable case. No polynomial time algorithms are presently known, and this would be a surprising and exciting breakthrough.

2. It is important to remember that the bounds given here are really *upper* bounds. The running time is exponential *iff* the splitting field is of exponential degree over k. If the degree of the splitting field is polynomially bounded, then so is the running time of the algorithm. Recall also that nearly all of the bounds follow from the factoring bound given in [9], and that bound is almost certainly not optimal. Any improvement in that bound will lead to great improvements in our bounds. Perhaps even more important to remember is that what is theoretically slow may be practically fast, and vice versa.

3. Our algorithm does not specifically examine the issue of determining if the expression equals zero. (We answer it, of course, in computing the minimal polynomial for α over k, but not efficiently.) Examining it leads to some interesting open questions on linear combinations of nested radicals. The case of nested square roots was treated by Borodin et al [3].

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